B. Tech. (3rd Sem)

(Common for CSE, CSE with Specialization in Data Science, CSE with specialization in Cloud Technology & Information Security)

BCSE-504 (Object Oriented Programming using JAVA)

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3	0	0	**Continuous evaluation**	**40**
			End semester exam	**60**
			Total marks	**100**
			Credits	**3.0**

Course Objectives:

1. To introduce the fundamentals of Object-Oriented Programming and Java programming environment. To acquire knowledge about Java programming language.
2. To develop programming skills using classes, objects, inheritance, polymorphism, packages, and interfaces.
3. To familiarize students with exception handling and multithreading concepts for developing robust applications.
4. To enable students to design and develop GUI-based applications using AWT, and event handling techniques.

Unit:-1

Object Oriented Programming: Foundation to concept of Object oriented programming, Features of object oriented programming languages, Procedural Programming vs Object Oriented Programming, Advantages of Object Oriented Programming. History and features of Java language, Java Development Kit, Java Virtual Machine, Java API, Command line arguments, Visual Studio Code for Java programming.

Java Basics: Data Types, Variables and their Scope, Final Variables, Static variables, Operators, Control Flow-Block Scope, Loops, Conditional Statements, Break and Continue Statements, Arrays and string class, garbage collection.

Unit:-2

Core OOP Programming: Classes and Objects, Pillars of Object Oriented Programming, Methods, Constructors, Types of constructors, Access qualifiers.

Inheritance: Inheritance concept, Benefit of Inheritance, Types of inheritance, Using super and final with inheritance, Polymorphism and its type : Method overriding & method Overloading , Abstract classes.

Packages and Interface: Defining a Package, Access protection, importing packages, defining an interface, implementing interfaces, Extending interface.

Unit:-3

Exception Handling: Types of Exceptions, Handling of Exception using try, catch and finally, Multiple catch clauses, Nested try statements, throw and throws, Built-in exceptions, User defined exceptions, Checked and unchecked exceptions.

Multithreading: Differences between thread-based multitasking and process-based multitasking. Thread life cycle, Types of creating threads, thread priorities, synchronizing threads, inter thread communication.

Unit:-4

GUI Programming – AWT classes, working with Frame window, Graphics class, AWT controls, layout managers, Designing menus and dialog boxes. Swings.

Event Handling: Delegation Event Model, Event classes and event listener interfaces, Handling mouse and keyboard events, Adapter classes, Inner classes.

Advanced OOP Concept and Design Applications: SOLID Principles-Single Responsibility, Open-Closed Principle

Case Study: 1. Learning Management System (LMS) – Design and development of a Learning Management System using Java to manage students, instructors, courses, enrollments, assignments, and assessments. The case study demonstrates the application of Object-Oriented Programming concepts such as encapsulation, inheritance, polymorphism, and abstraction, along with interfaces, packages, exception handling, collections framework, and GUI development using AWT and Swing. It provides practical exposure to designing user-friendly and industry-oriented software applications using Java.

1. **Online Shopping System (E-Commerce Application) –** Analysis and implementation of an Online Shopping System using Java for managing customers, products, shopping carts, and order processing. The case study emphasizes the

application of Object-Oriented Programming principles including encapsulation, inheritance, polymorphism, and abstraction, along with interfaces, packages, exception handling, collections framework, and GUI development using AWT and Swing. It enables students to gain practical experience in developing interactive and object-oriented software applications.

Course Outcomes: After completion of this course, students will be able to:

1. Apply Java programming fundamentals and object-oriented concepts to develop simple Java applications.
2. Design object-oriented solutions using classes, inheritance, polymorphism, packages, and interfaces.
3. Implement exception handling and multithreading techniques in Java programs.
4. Develop GUI-based applications using AWT components, and event-driven programming.

Instructions for paper setter: All Questions are compulsory. The Question paper is divided in to four sections A, B, C and D. Section A is compulsory and comprises of 12 questions of one mark each, 3 from each unit. The questions shall be asked in such a manner that there are no direct answers including one word answer, fill in the blanks or multiple choice questions. Section B comprises of 4 questions of 2 marks each, one from each unit. Section C Comprises of 4 questions of 4 marks each, one from each unit. Section D

Comprises of 4 questions of 6 marks each, one from each unit. There is no overall choice, however internal choice may be provided in section C and D, if paper setter so desires.

Text Books:

1. "Core Java Volume-I and II" 2nd edition-Sun Micro System.
2. Java: The Complete Reference, Herbert Schildt, McGraw Hill Education.

Reference Books:

1. Programming with Java, E. Balaguruswamy, McGraw Hill Education.
2. Rick Dranell , "HTML 4 Unleashed", Second edition, Tech media publication.
3. Head First Java, O'Reilly Media.
4. Object-Oriented Programming with Java, McGraw Hill Education.
5. Java How to Program, Pearson Education.